of roughly the same shape, these bounds go up and down at different rates. Further, since the data point maximizing this bound is the point that UCB selects to query for optimization, different bounds, that is, different values of β, will induce different optimization behaviors.

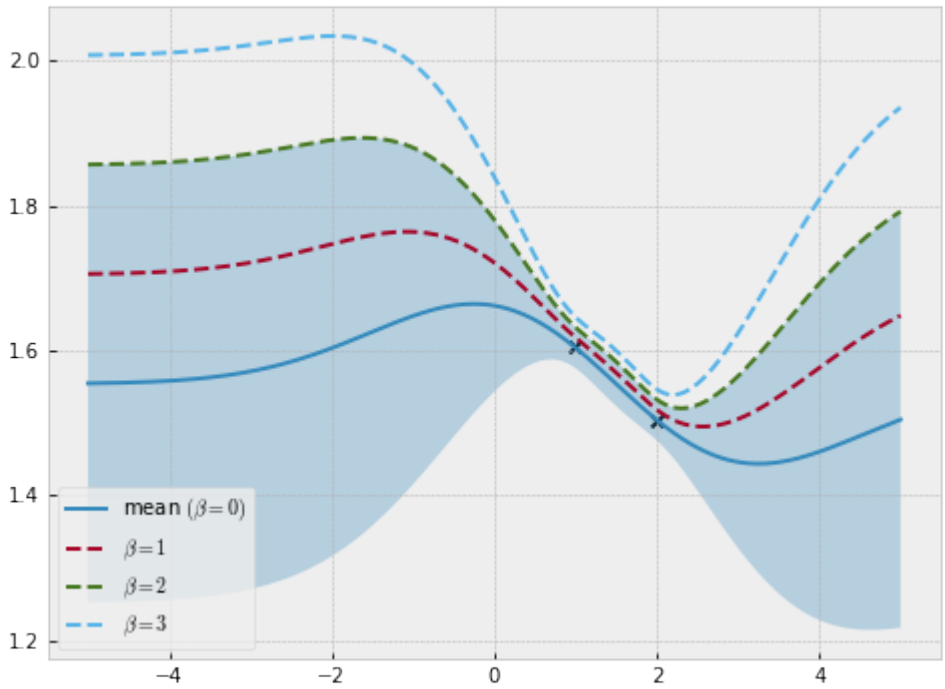


Figure 5.8 Different Upper Confidence Bounds of a Gaussian process, corresponding to different credible intervals and values. The larger is, the more explorative the UCB policy becomes.

IMPORTANT **How the parameter controls exploration/exploitation**

The smaller is, the more exploitative UCB becomes. Conversely, the larger is, the more explorative UCB becomes.

We can see that the value of β controls the behavior of UCB by inspecting the formula for the acquisition score: $\mu + \beta\sigma$. When β is small, the mean μ contributes the most to the acquisition score. So, the data point with the highest predictive mean will maximize this score. This selection corresponds to pure exploitation, as we are simply picking out the point with the largest predicted value. When β is large, on the other hand, the standard deviation σ, which quantifies our uncertainty, becomes more important in the UCB score, emphasizing the need for exploration.

We see this distinction in Figure 5.8, where around 0 is where we achieve the highest predictive mean, so that is the region of exploitation. As β increases, the point at which the different upper bounds peak gradually move to the left where there is more uncertainty in our prediction.

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